

PROJECT 02

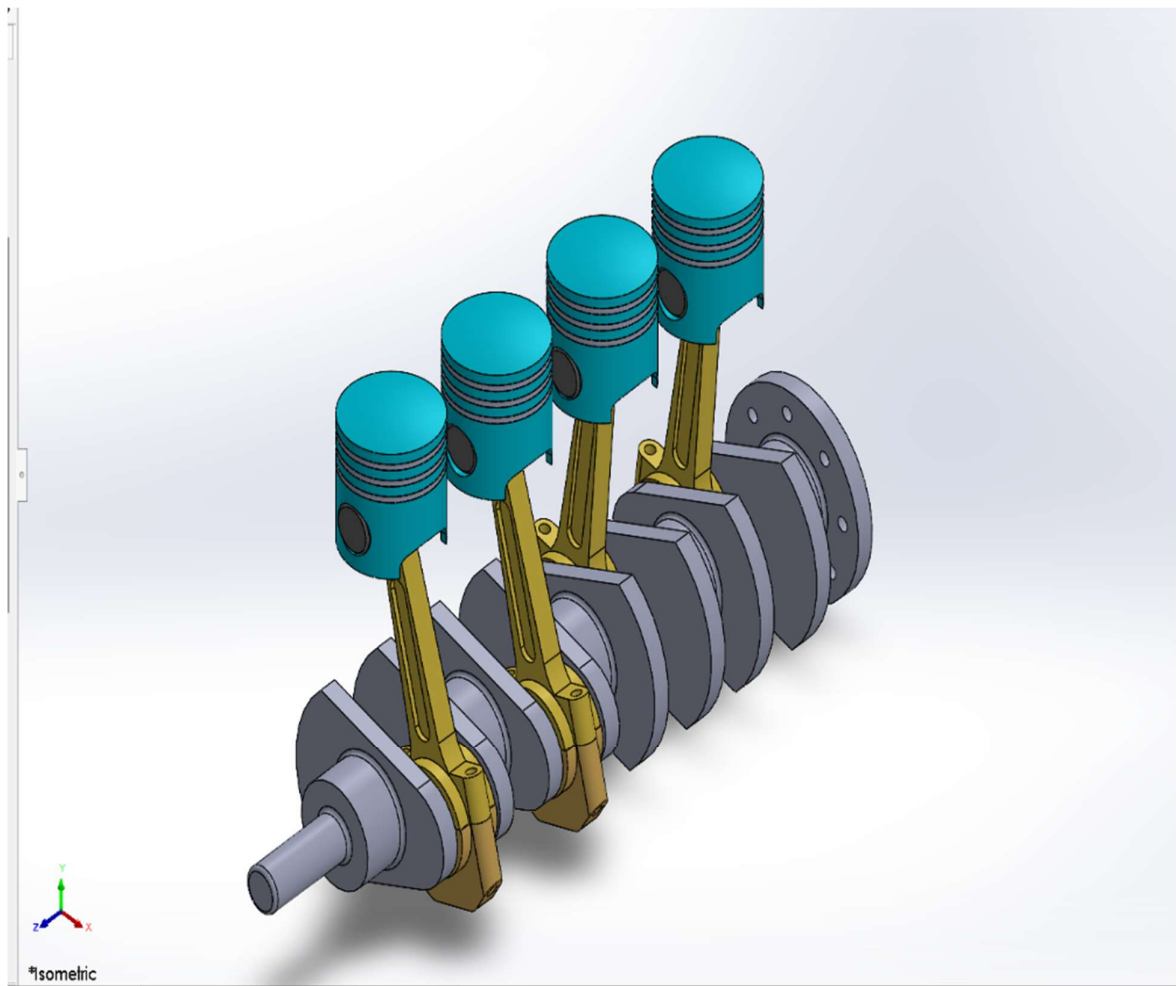
Inline Four-Cylinder Crankshaft Assembly Design

Software Used: SolidWorks Premium 2021

Project Type: Mechanical Assembly Design

Category: CAD Design | Mechanical Engineering | Motion Study

Author: Govind Goyal

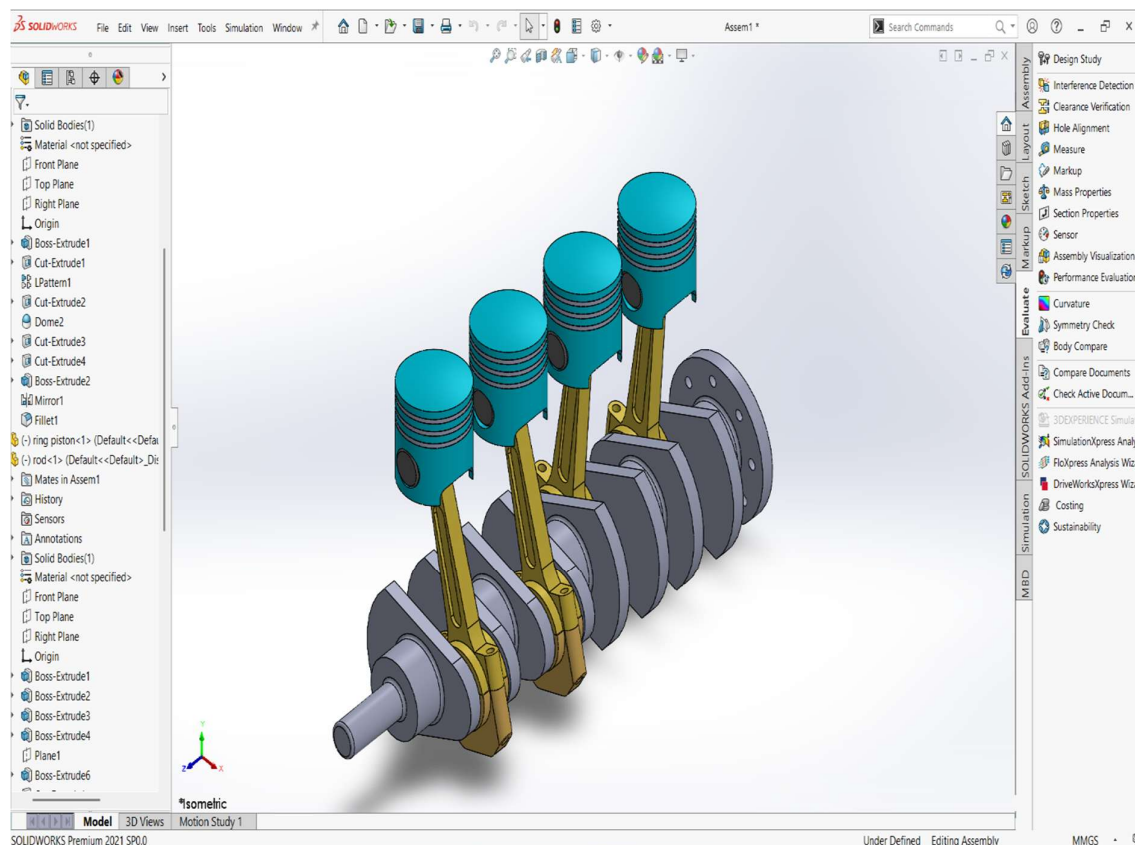


1. Project Overview

The Inline Four-Cylinder Crankshaft Assembly is a mechanical CAD project developed in SolidWorks Premium 2021 to understand the working principle of an internal combustion engine's crank-slider mechanism.

The project involves designing individual engine components and assembling them into a complete mechanism capable of demonstrating synchronized piston motion. The assembly illustrates how the rotational motion of the crankshaft is converted into the reciprocating motion of the pistons through connecting rods.

The primary objective of this project was to strengthen CAD modeling skills, improve assembly design techniques, and gain practical knowledge of engine kinematics while creating a professional engineering portfolio project.



2. Project Objectives

- Design an Inline Four-Cylinder Crankshaft Assembly.
- Model each engine component individually.
- Create a fully constrained mechanical assembly.
- Demonstrate piston motion using SolidWorks Motion Study.
- Understand crank-slider mechanism and engine kinematics.
- Develop a professional engineering portfolio project.
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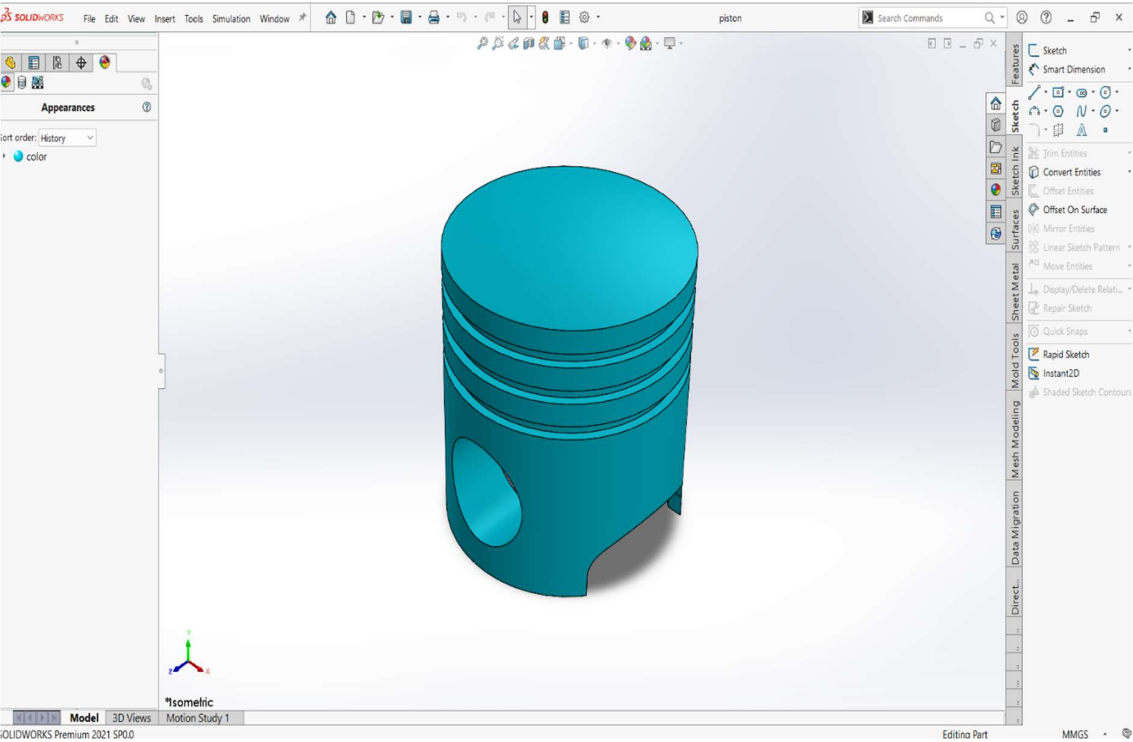
3. Software Used

Software	Purpose
SolidWorks Premium 2021	3D Part Modeling
SolidWorks Assembly	Assembly Design
SolidWorks Motion Study	Motion Simulation
SolidWorks Drawing	Engineering Drawings

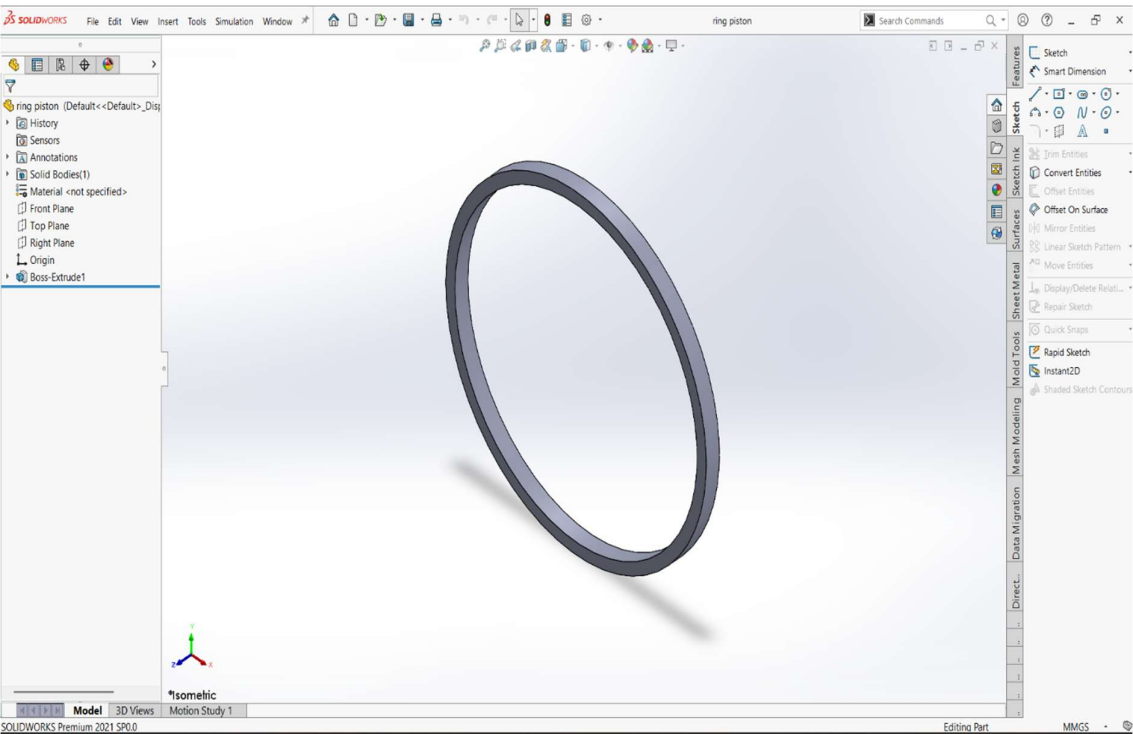
4. Components Designed

Component	Quantity	Material
Crankshaft	1	Alloy Steel
Connecting Rod	4	Alloy Steel
Pistons	4	Aluminum Alloy

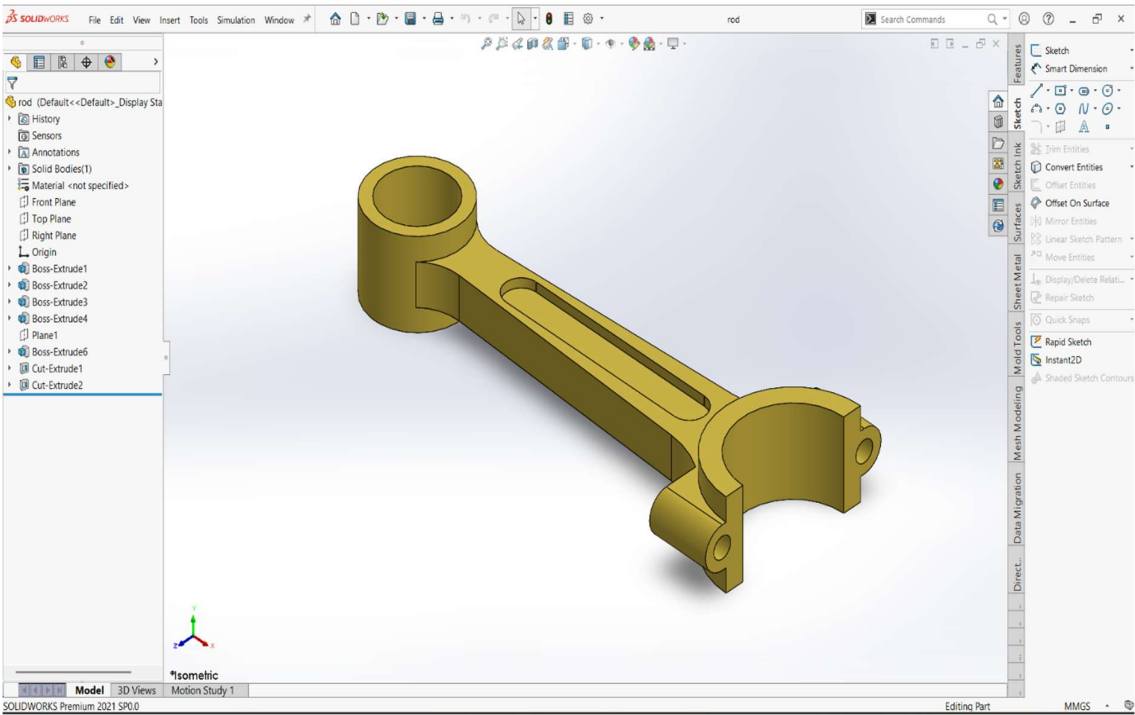
PISTON



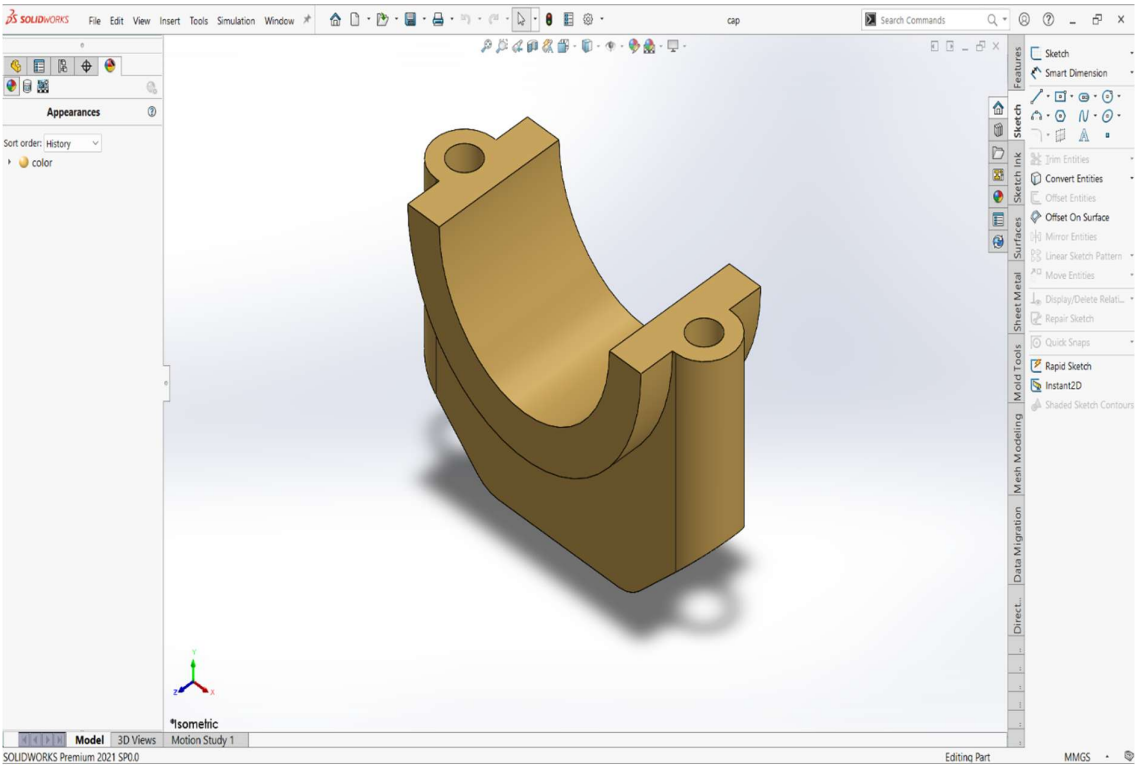
PISTON RING



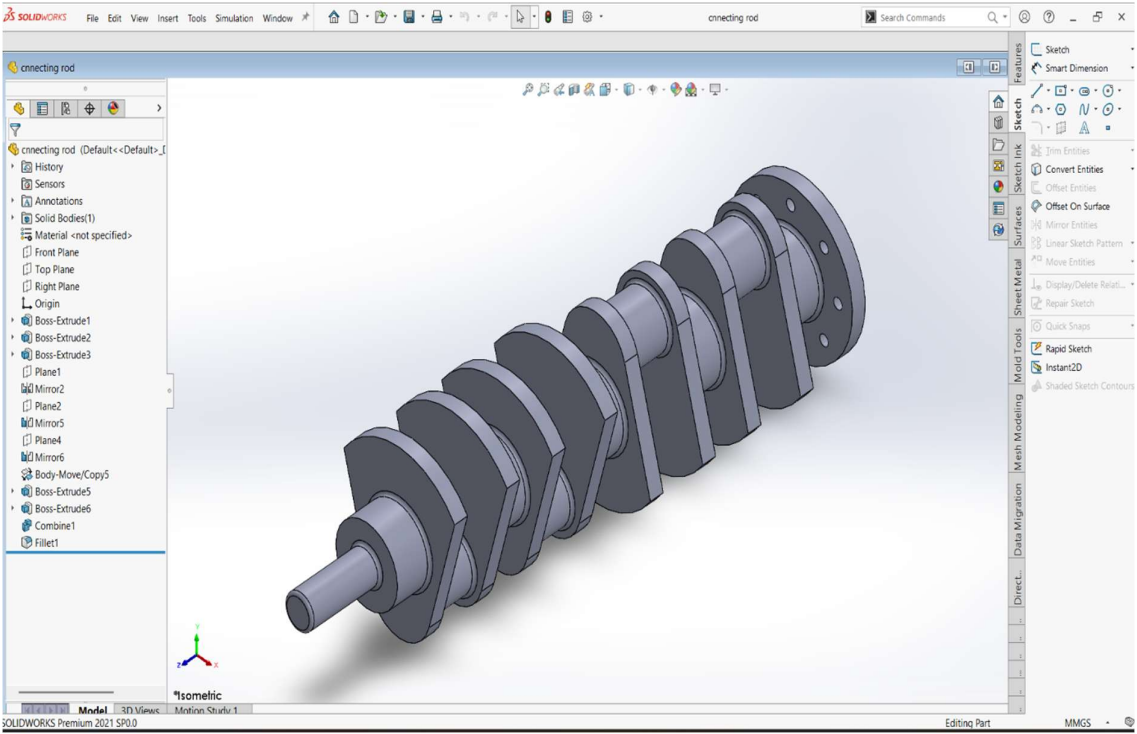
CONNECTING ROD



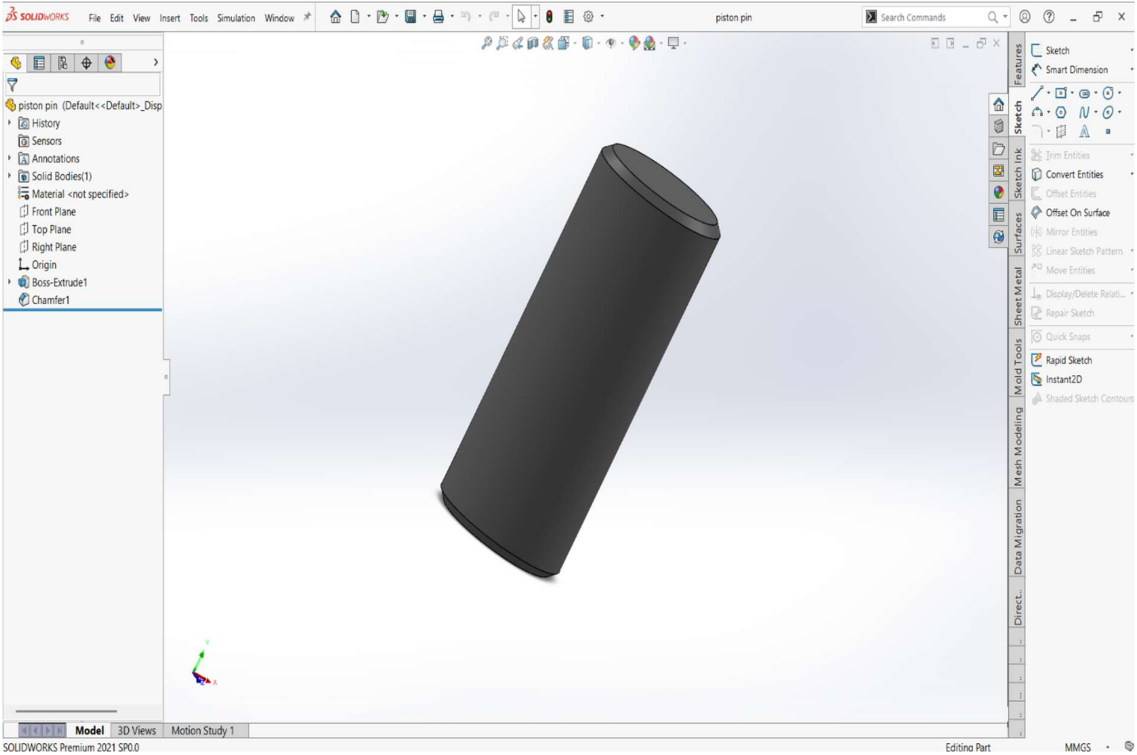
CONNECTING ROD CAP



CRANKSHAFT



PISTON PIN



5. Material Selection

Crankshaft

Material: Alloy Steel

Reason

- High strength
- Excellent fatigue resistance
- Suitable for rotating components
- High torsional rigidity

Connecting Rod

Material: Alloy Steel

Reason

- High tensile strength
- Excellent fatigue life
- Suitable for cyclic loading
- Durable under engine operating conditions

Pistons

Material: Aluminum Alloy

Reason

- Lightweight
- Good thermal conductivity
- Reduced reciprocating mass
- Improved engine efficiency

6. Assembly Features

The assembly was created using standard SolidWorks mates to accurately replicate the mechanical motion of an inline four-cylinder engine.

Features

- Concentric Mates
- Coincident Mates
- Mechanical Alignment
- Fully Constrained Assembly
- Rotational Crankshaft
- Synchronized Piston Motion

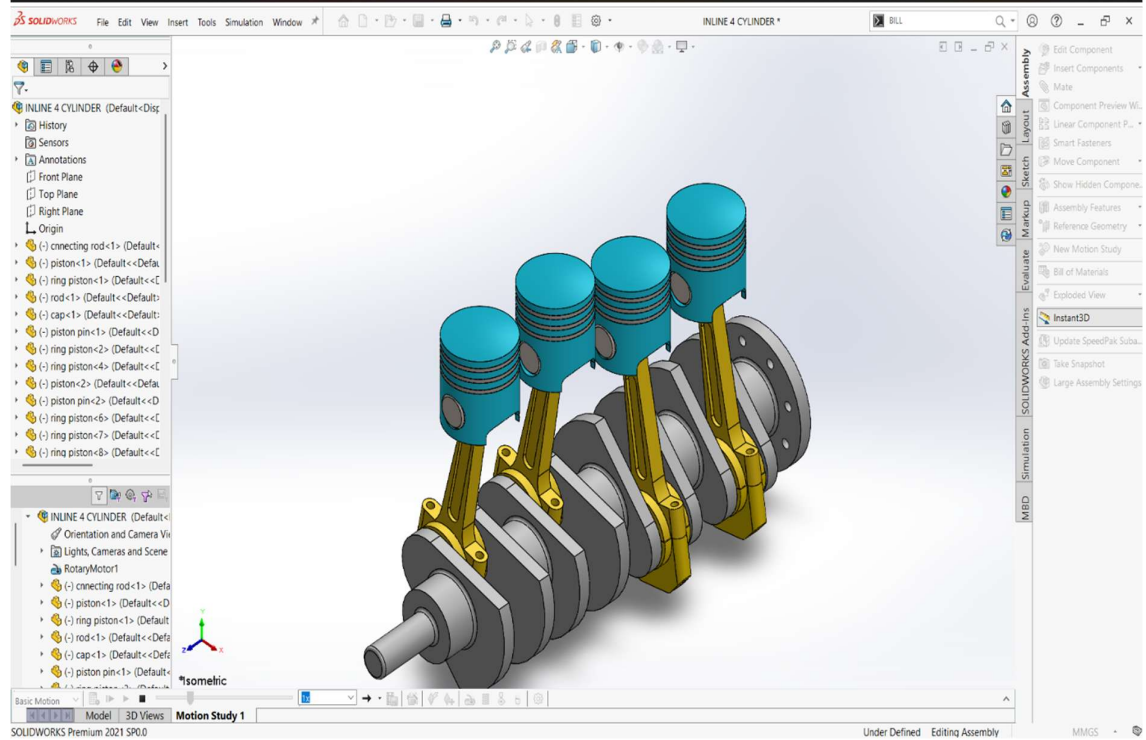
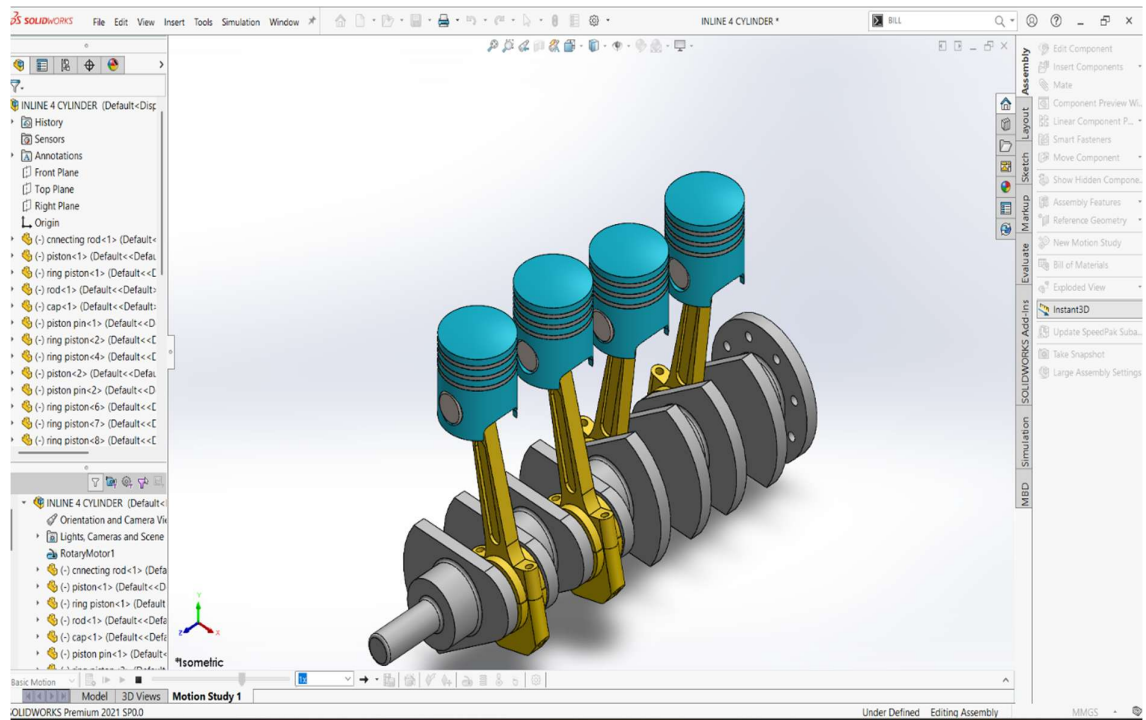
7. Motion Study

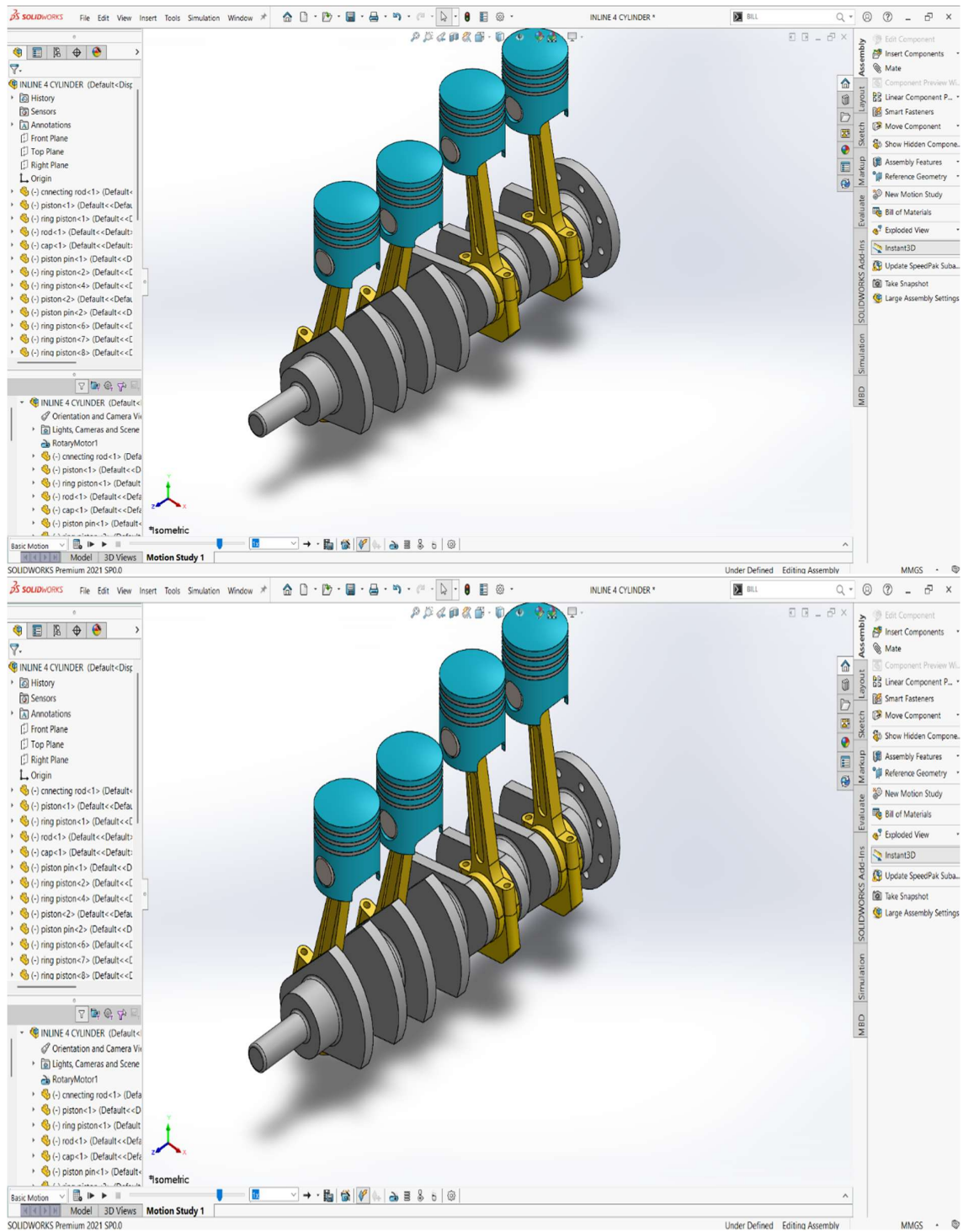
The completed assembly was validated using SolidWorks Motion Study.

The animation demonstrates:

- Continuous crankshaft rotation
- Connecting rod oscillation
- Piston reciprocating motion
- Proper synchronization of all four cylinders

This verification ensures that the assembly behaves according to the intended crank-slider mechanism.



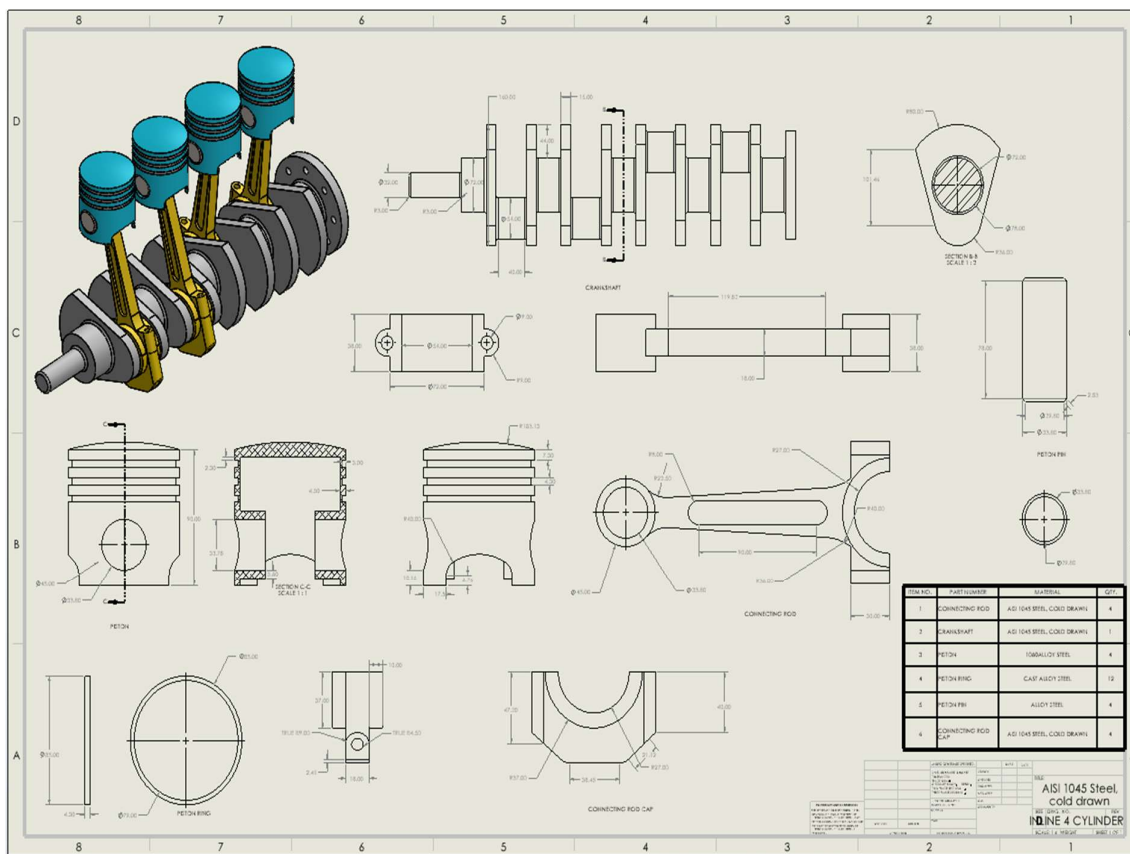


8. Engineering Drawings

Engineering drawings were prepared for the designed components using SolidWorks Drawing.

The drawings include:

- Orthographic Views
- Isometric Views
- Dimensions
- Centerlines
- Section Views



9. Engineering Concepts Used

- Crank-Slider Mechanism
- Rotary Motion
- Reciprocating Motion
- Mechanical Assembly Design
- Motion Transmission
- Mechanical Constraints
- Engineering Drawings

10. Skills Demonstrated

CAD Design

- Part Modeling
- Assembly Design
- Feature-Based Modeling

Mechanical Engineering

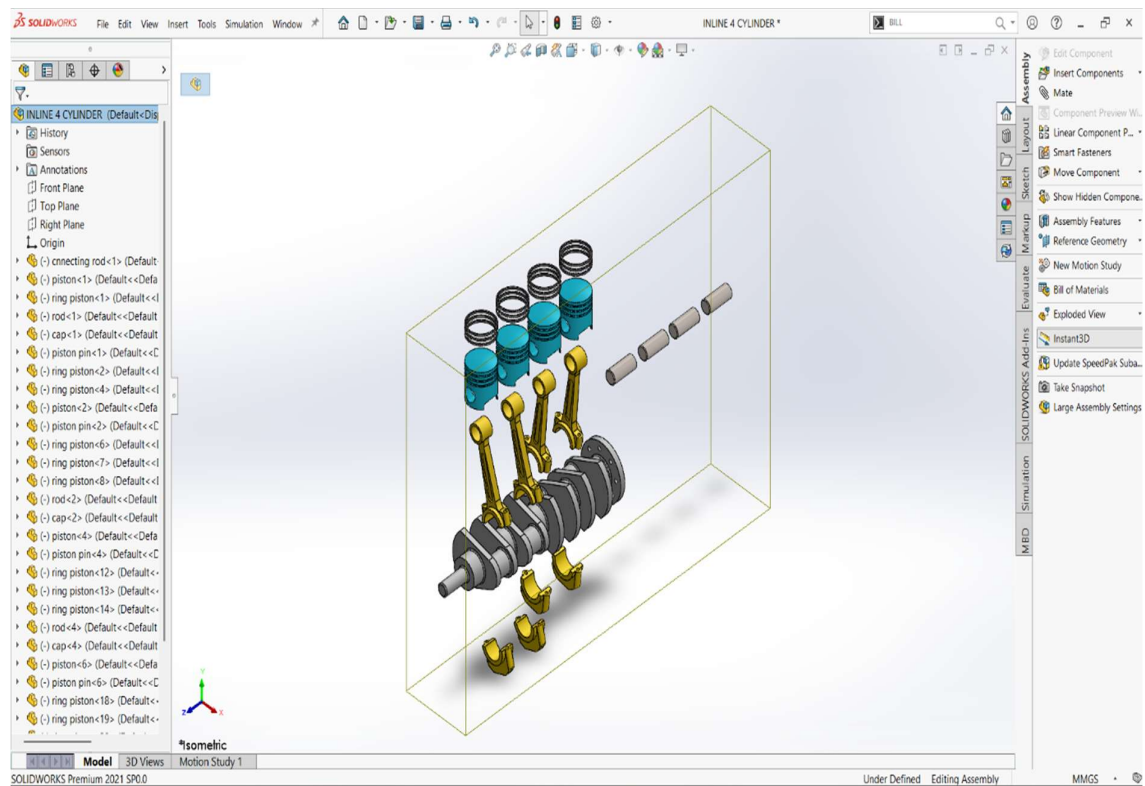
- Engine Mechanism
- Motion Analysis
- Mechanical Design

Documentation

- Engineering Drawings
- Technical Documentation
- Portfolio Development

11. Deliverables

- i. 3D CAD Parts
- ii. Complete Assembly
- iii. Motion Study Animation
- iv. Engineering Drawings
- v. High-Quality Render
- vi. Portfolio PDF



THANK YOU